A Study of Stochastic Sequence Generation: Comparing LSTM and RAG

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Abstract

Jazz has spontaneous features, including the player's emotion, audiences, and improvisation. Jazz players especially copy and practice for soloing. However, a jazz solo requires considerable effort: listening, harmony analysis, and recreating. According to these features, we suggest a jazz-lick generator, leveraging AI-driven approaches: Long Short-Term Memory and Retrieval-Augmented Generation. These approaches are widely pondered to recreate the licks. This study aims to develop jazz lick generation (stochastic sequence), leveraging a dataset of licks. Consequently, we compare two methods: a bidirectional LSTM and a pitch-interval RAG. Retrieval-augmented approaches show improved results. Specifically, the pitch-interval RAG model was evaluated, repeatedly varying the alpha weights. It tends to present a considerably higher hit rate than the Bi-LSTM. The high hit rate exhibited preservation of melodic and harmonic aspects, suggesting remarkable features of the jazz language; furthermore, this indicates that interval similarity demonstrates the spontaneous context of the jazz language.

Keywords: LSTM, RAG, Jazz licks, Music Generation, Jazz Improvisation

1 Introduction

Jazz predominantly has colorful characteristics, such as a complex musical process, various harmonics with chord changes, melodic contours, spontaneous solos known as 'licks', and unpredictable variables. Of these, jazz musicians, especially, agonize to enhance the improvisational jazz licks, routinely copying (transcription of recordings) and practicing (melodic patterns used by well-known performers) the short melodic phrases. In real jazz performances (known as 'gigs'[1]), jazz musicians appropriately harmonize and transform such studied licks and patterns to recreate spontaneous

improvisations. However, recreating new jazz licks requires considerable time and persistent effort; furthermore, jazz performers, even well-known, tend to reuse familiar patterns (paradoxically, it occasionally may make the unique sound-color). Addressing these challenges, this study practically suggests an efficient system, ultimately generating new jazz licks for jazz musicians. This study, promisingly, aims to be beneficial for both practice and real jazz performance.

Bebop (an intricate and fast tendency) and hard bop (a melodic and fast tendency) styles in jazz improvisation, in particular, exhibit unique characteristics: eighth-note sequences (suitable for computational modeling with no rests), including wide-range tempos (200 to 350 BPM), passing tones, chromatic approaches, and altered tensions. Furthermore, those improvisations tend to follow certain rhythmic and harmonic patterns: Specifically, chord tones are mainly placed on the downbeats (1 and 3) to emphasize the harmonic structure; Oppositely passing tones and tensions are placed on the upbeats (2 and 4) for smooth melodic flow. These main properties allow licks to be represented effectively as sequential data, making them mostly optimizable for machine learning approaches. Through these diverse features, we aim to develop a jazz-lick generator system; it utilizes various AI approaches such as Long Short-Term Memory (LSTM), Retrieval-Augmented Generation (RAG), and Large Language Models (LLMs). LSTM is frequently used for modeling sequential features in musical data. However, the sequence-based models, depending on learned parameters, often fail to generate the licks due to limited contextual understanding. In contrast, RAG has recently gained attention as an initiative for enhancing generative models, retrieving relevant examples from a dataset; it presents recreated data, consequently enriching the model's expressiveness. In this study, we investigate the potential possibility of LSTM and RAG. We construct a dataset of bebop lick patterns extracted from symbolic MIDI data, thoroughly evaluating two methods: (1) a bidirectional LSTM, (2) a pitch-interval RAG. The goal of this study is to examine: different musical similarity and its influence on generated jazz licks. Furthermore, this work supports future studies with Transformer models.

Contributions. The main contributions of this study are as follows:

- A jazz improvisation system including retrieval-augmented generation with melodic similarity measures.
- A comparative analysis of LSTM and Pitch-interval based retrieval strategies for lick generation.
- A delicate evaluation demonstrating how interval similarity preserves melodic contour in jazz licks.

2 Related Work

Data-driven music generation has been widely studied using AI-based techniques, specifically through rule-based and early data-driven approaches. The SaxEx system analyzes recorded saxophone performances to extract data and model expressive parameters, applying the inferred transformations to new phrases [2]. Furthermore, Pachet's research has yielded the 'surprise' in jazz chord progressions: recurring patterns, chord substitution rules, and models of harmonic surprise [3].

Since the 2010s, various studies leveraging machine learning have been conducted. For instance, Gillick proposed various approaches for generating solos, mainly using slopes and Markov chains [4]. Subsequently, DeepBach [5] was a graphical model that aimed at modeling polyphonic music (hymn-like pieces) through pseudo-Gibbs sampling; it also provides the MuseScore plugin. Around this time, MuseGan [6] proposed three models (the jamming model, the composer model, and the hybrid model) for multi-track music generation with generative adversarial networks (GANs), providing the additional tracks generated by given notes composed by humans. In parallel, Choi studied a ResNet/LSTM-based model that offers the regularity and inconsistency of forecasting data [7]. Soon after, Huang proposed an algorithm that reduces the intermediate memory for Transformer, fundamentally enabling it to generate minute-long compositions and accompaniments conditioned on melodies [8].

In addition, a study [9] proposed an algorithm of Long Short-Term Memory (LSTM) for generating musical notes, utilizing MIDI file format. The model mainly learned the sequences of polyphonic music and successfully produced new melodies. Similarly, Shah aimed to generate musical notes through a given input sequence, producing musical continuous and melodious sequences; it proposes new methods for data ingestion on Long Short-Term Memory (LSTM) based models, leveraging Google Magenta's inbuilt models [10]. In a similar vein, Zhao designed a novel system for emotional music generation with various approaches: Biaxial LSTM (to generate polyphonic music), LookBack (the architecture to improve the long-term structure), steerable parameters (for 4 basic emotions divided by Russell's VA Space) [11].

More recently, AI-driven approaches have been developing, such as Transformer and RAG. Briot studied musical content generation using deep learning through five dimensions: Objective, Representation, Architecture, Challenge, and Strategy; he proposed various models and techniques (with tentative multidimensional approaches) [12]. Moreover, Gunawan studied an automatic music generator with a MIDI dataset, using long short-term memory (LSTM) and gated recurrent units (GRUs) networks; the result shows the layer GRU model performs better to resemble the composer pattern [13]. At this point, Hakimi addressed the generation of personalized music that learns and optimizes user-specific musical taste, using BebopNet (a model trained on a corpus of jazz improvisations) that generates improvisations based on chord progression [14]. Meanwhile, Wu studied the Jazz Transformer (a generative model), utilizing the Transformer-XL (a neural sequence model) for modeling sheet music of Jazz [15]. On the other hand, Lewis suggested two RAG formulations: one which conditions on the same retrieved passages across the whole generated sequence, and the other which can use different passages per token for a general-purpose fine-tuning [16].

In 2021, Madaghiele introduced MINGUS (a Transformer-based architecture for modelling and generating monophonic jazz melodic lines) for automatic music generation, using Sequence to Sequence (Seq2Seq) approaches [17]. Likewise, Conner provided a brief intuition, theory, and application of LSTMs in music generation, develops the network, addresses issues and challenges faced, and suggests potential future improvements [18]. At the same time, Rafraf presented various approaches for Music Generation in LSTM Networks, leveraging cosine similarity, interval methods, and

more [19]. There have also been studies on polyphonic harmony through deep learning [20]. Additionally, Copet introduced MusicGen (a single Language Model (LM) that operates over streams of music representation) which is comprised of a single-stage transformer LM together with efficient token interleaving patterns; it demonstrates the result of generating high-quality samples [21].

This study has explored to develop a system of novel and spontaneous jazz licks generating. Furthermore, our approach fundamentally yields the exciting challenges and future works of modeling jazz improvisation.

3 Method

3.1 Dataset Construction

In this study, we utilize datasets typically extracted from the Weimar Jazz Database (WJazzD v1.4.1, [22]); these were reconfigured and formed into 2,271 melodic patterns. Each pattern is mapped to jazz licks based on a dominant seventh chord (in a 2-5-1 chord progression), composing one bar of eighth notes (melodic segment). Continuous eighth note licks (densely almost no rest) as jazz licks (as Bebop and Hard-bop style) are desirable, mathematically, to calculate and train. This rhythmic feature of the mapped patterns is simplifiable for the datasets for training and evaluation. It enables only melodic phrases to be modeled as pitch sequences (exclusive 8th-notes with no rests), regardless of rhythm. In addition, jazz improvisation frequently tends to use various characteristics (with tension notes) scale structures: the altered scale, the diminished scale, the whole tone scale, the bebop dominant 7 scale, among others. Given that, the jazz lick, appropriate domain, is effective for studying algorithmic improvisation generation. The dataset was precisely converted into input-output pairs (total four notes) consisting of three-note (input sequences) and another one-note (target note) by using a sliding window approach: $[P_1, P_2, P_3] \rightarrow P_4$. Furthermore, the note range is 2 octaves, encoding 26 notes (C4 C6) in a row. The main principle is to reuse the next input from the previous predicted result:

- **using the init input:** input the 3 notes to the model.
- **prediction and extends:** analyzing the 3 input notes and predicting the next one note.
- **feedback loop:** one predicted note is added and extends the sequence.

This process repeats 4 times, where the initial 3 notes generate the 8 note (one bar).

3.2 Bi-LSTM Sequence Model

The Bidirectional Long Short-Term Memory (Bi-LSTM) is employed as a baseline model and designed to predict the next pitch given a sequence of three input notes. This model precisely implements a 3-to-1(many-to-one method approach) sequence-to-note prediction structure using Keras (TensorFlow's high-level API).

3.2.1 Model Architecture

The model consists of the following layers:

- **Bidirectional LSTM layer:** 512 units, which enables the model to capture sequential dependencies in both forward and backward directions (simultaneously).
- **Dropout layer:** 0.3, applied to prevent overfitting.
- **LSTM layer:** 128 units, encoding the temporal representation.
- **Dense output layer:** Softmax activation function (for the most likely next pitch note) to produce a probability distribution over all possible pitches (total number of pitch notes: 26).

The input to the model has the shape $(\texttt{INPUT_LENGTH}, 1)$, where $\texttt{INPUT_LENGTH} = 3$ for the three-note input window.

$$\text{Input} = \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix} \in \mathbb{R}^{3\times 1} \longrightarrow \text{Target} = [n_4] \tag{1}$$

Figure 1 shows the architecture of the Bi-LSTM sequence model.

$$\begin{aligned} \text{Input (3-note)} &\rightarrow \text{Bi-LSTM (512)} \\ &\rightarrow \text{Dropout (0.3)} \\ &\rightarrow \text{LSTM (128)} \\ &\rightarrow \text{Dropout (0.3)} \\ &\rightarrow \text{Dense (Softmax, 26)} \end{aligned}$$

Fig. 1 Architecture of the Bi-LSTM sequence model for predicting the next note in jazz lick generation. The input is a three-note pitch sequence, processed by a Bidirectional LSTM layer, followed by Dropout and a Dense output layer with softmax activation.

3.2.2 Hyperparameters and Training

The model (Bi-LSTM) is repeatedly trained using the Adam optimizer and the categorical cross-entropy loss function. Table 1 indicates the key hyperparameters summarized below.

3.2.3 Evaluation Metrics

The performance of the Bi-LSTM model is evaluated using:

- **Hit Rate:** percentage of predicted notes matching the target note.
- **Interval Variety:** mean interval difference between consecutive generated notes.
- **Scale Fit:** percentage of notes conforming to the target harmonic scale.

Table 1 Stacked Bi-LSTM model hyperparameters

Hyperparameter	**Value**
First LSTM units (Bi-LSTM)	512
Second LSTM units	128
Bidirectional	Yes (first layer only)
Number of LSTM layers	2
Dropout rate	0.3
Batch size	32
Learning rate	0.001 (Adam default)
Optimizer	Adam
Loss function	Categorical cross-entropy
Epochs	50
Input sequence length	3
Output dimension	26
Activation (output)	Softmax

These metrics show how well the model (creatively and spontaneously) yields sequential dependencies and melodic characteristics (contours) of jazz licks.

3.3 Pitch-interval Retrieval-Augmented Generation

In this section, we propose the pitch-interval retrieval model approach, incorporating calculated data with both absolute pitch notes and intervals.

Each melodic row in dataset (already mapped) is formalized as a sequence of three notes, represented by a Pitch Vector (P) and an Interval Vector (I):

$$P = [P_1, P_2, P_3] \tag{2}$$

$$I = [P_2 - P_1, P_3 - P_2] \tag{3}$$

For example, a melodic sequence of C4, E4, and G4 (MIDI notes 60, 64, 67) results in a Pitch Vector $P = [60, 64, 67]$ and an Interval Vector $I = [4, 3]$ (specifically calculated as $64 - 60 = 4$ and $67 - 64 = 3$).
As shown in Table 2, the **Pitch Vector (P)** yields the absolute height; furthermore the **Interval Vector (I)** allows for interval patterns (transposition manageable). This shows the same melodic pattern in different melody pitch-notes (licks).

Given an input (three notes randomly or returned), this system (jazz lick generator) retrieves the most similar melodic patterns from the dataset based on the cosine similarity approach; It is, significantly, utilized for jazz lick generation as follows: for each result, the cosine similarity of pitch and interval is calculated, varying the α parameter.

Table 2 Example of Pitch and Interval Vectors

Melody (Notes)	**Pitch Vector (P)**	**Interval Vector (I)**
C4 – E4 – G4 (C Major Triad)	$[60, 64, 67]$	$[4, 3]$
D4 – F#4 – A4 (D Major Triad)	$[62, 66, 69]$	$[4, 3]$

$$Score = \alpha \cdot Similarity_{interval} + (1 - \alpha) \cdot Similarity_{pitch} \tag{4}$$

By varying the α parameter, the system tends to recreate either the melodic contour or the specific pitch note, effectively converting the same jazz licks across different keys or the same dominant 7th chord. Depending on the α parameter, generated jazz lick affects the creativity and originality of tendency.

4 Experiments

Two generative models (Bi-LSTM, Pitch-Interval RAG) were precisely evaluated to compare their performance in predicting jazz lick sequences. This study focuses on these two models for developing an LSTM and RAG web application. The models are summarized in the following Table 3:

- **Bi-LSTM**: Sequence-based pitch prediction model.
- **Pitch-Interval RAG**: Retrieval-augmented generation using pitch-interval similarity.

Generated melodic sequences (jazz licks) were evaluated considering the following set of criteria:

- **Hit Rate**: The percentage (%) of generated notes matching the target notes in the dataset.
- **Interval Variety**: The average interval change between consecutive notes (within one-bar jazz licks), measuring melodic diversity.
- **Scale Fit**: Degree to which generated notes conform to dominant 7th scale contexts, including altered, whole-tone, and diminished scales, among others.

$$\text{Scale Fit (\%)} = \frac{\text{\# notes in target scales}}{\text{Total generated notes}} \times 100 \tag{5}$$

4.1 Detailed Analysis

- **Bi-LSTM:**
  The Bi-LSTM model yielded sequences with a Hit Rate of 10% and an average interval variety of 3.98; It shows a (relatively) low Hit Rate, meaning the model tends

Table 3 Comparison of generation performance metrics.

Model	Hit Rate (%)	Interval Variety	Scale Fit (%)
Bi-LSTM	10	3.98	58.7
Pitch-Interval RAG	50	4	66

to generate licks with creative contours. Scale analysis indicates proper alignment (predominantly) with dominant 7th scale (Mixolydian scale) contexts:

- Chord Tone: 46%
- Altered Tension: 18%
- Natural Tension: 6%
- Altered Scale Fit: 66%
- Whole-tone Scale Fit: 50%
- Diminished Scale Fit: 60%

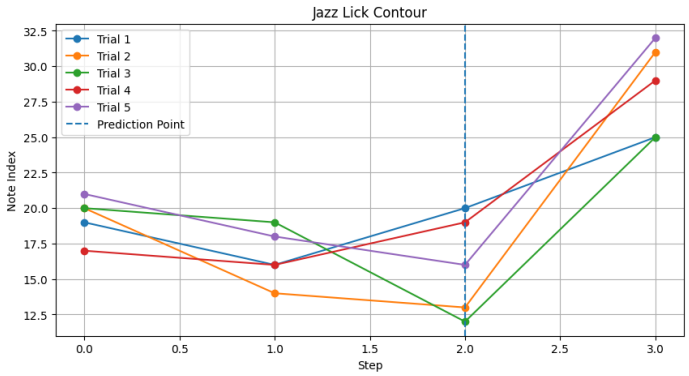


Fig. 2 Melodic Contour Analysis of Bi-LSTM: Visualization of the transition from the input to the predicted note.

As shown in Figure 2, the Bi-LSTM model demonstrates the highest similarity to chord tones (46%) while maintaining a (fairly) consistent melodic contour, whereas the diminished scale (60%) presents the tension-release/resolution tendency (typically) of dominant chord contexts in jazz.

- **Pitch-interval RAG:** The pitch-interval retrieval model achieved a Hit Rate of 50.0% and an interval variety of 4, suggesting more conservative melodic transitions. These results present various aspects, such as predictable contours and stable intervals; it is practically applicable for jazz gigs. Scale fit also improved to 66%, with real-time C7 Dominant analysis following below:

– Chord Tone: 40%
– Altered Tension: 34%
– Natural Tension: 4%
– Altered Scale Fit: 78%
– Whole-tone Scale Fit: 44%
– Diminished Scale Fit: 76%

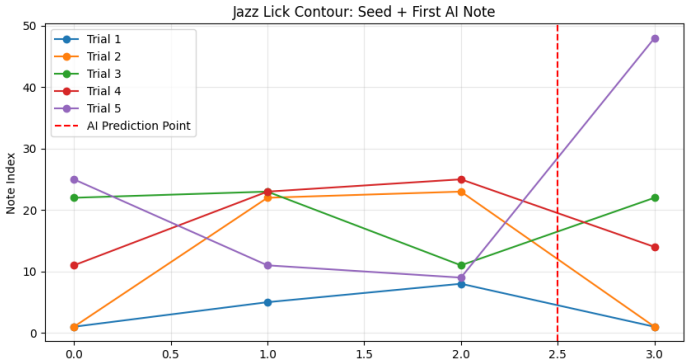


Fig. 3 Melodic Contour Analysis of Pitch-interval RAG: Visualization of the transition from the input to the predicted note.

As illustrated in Figure 3, the Pitch-interval RAG demonstrates a more balanced distribution of tensions (in chord/scale), compared to the Bi-LSTM. Even though chord tones are lower (40%), the remarkably higher Altered Scale Fit (78%) — properly—indicates the jazz context from the database.

This experiment results show that retrieval-augmented generation improves—creatively—musical approaches and jazz lick diversity (spontaneously) compared to the baseline Bi-LSTM; especially, interval-based similarity data typically tend to preserve the melodic contour.

5 Results

Based on the experimental results (Table 3), the RAG model yields remarkable characteristics compared to the Bi-LSTM approach. Specifically, the pitch-interval RAG model achieved a superior Hit Rate (50%); conversely, compared to the Bi-LSTM baseline (10%), it maintained a similar level of interval variety (4 compared to 3.98). These results show that the pitch-interval approach (including melodic shape) is more effective for creating jazz licks.

6 Conclusion

This study aimed to develop, ultimately, a jazz lick generation system, thoroughly analyzing a dataset of dominant seventh jazz licks using Bi-LSTM and retrieval-augmented

generation (RAG). Using the analyzed dataset, we relatively evaluated two configurations: a bidirectional Bi-LSTM baseline and a pitch-interval RAG model. This jazz lick generation system demonstrates that retrieval-augmented approaches magnificently outperform the features of generated jazz licks (as bebop and hardbop contours) compared to LSTM methods. The pitch-interval RAG model achieved the highest performance with a hit rate of roughly 50%, providing a stable mechanism for generating delicate melodic lines (licks) simultaneously. In contrast, LSTM methods tend to yield a slightly lower hit rate; it allows unexpected melodic contours and harmonic aspects. In this study, we found, interestingly, that pitch-interval retrieval approaches prioritize melodic contours which are more suitable to apply (realistically) in jazz performance. Given with Jazz lick suggestions by RAG, it encourages jazz musicians to expand their improvisational vocabulary and explore novel melodic possibilities and approaches through practice (memorizing and re-creating) and performance (for applying). Future work will involve scaling the system with larger symbolic datasets and developing an interactive web-based application (for Mobile/PC) for practical usage in jazz performance (gigs).
Model Summary: The Bi-LSTM tendentially serves as a stable baseline for local dependencies. Oppositely, the pitch-interval RAG, compatibly, offers creative notes and conventional melodic structures generation, varying parameters. Furthermore, it encourages ultimately flexible sequences that prioritize melodic integrity and structural contour.

Declarations

Conflict of Interest

The author declares no conflict of interest.

Ethical Approval

Not applicable.

Author Contribution

The author conducted the study, designed the methodology and wrote the manuscript.

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Availability of Data and Materials

The datasets used in this study are available from the corresponding author upon reasonable request.

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